Module 1 Homework 1 Challenge

Below is a scored review of your assessment. All questions are shown.

Earned 59,994 of 99,990 points. 60%

Main Question Set

Solve six of the first ten questions to achieve a passing score of 60%. Don't miss the learning opportunity in the rest of the problems, though -- spend the full time and attempt every one of them!

Correct Answer Partially Correct Incorrect Answer

1 Which of these fractions is the biggest?

Your Answer $\frac{5}{6}$

Correct Answer $\frac{5}{6}$

Explanation

Employing a basketball analogy, like the example from Day 4 in the videos, can help us to see which of these fractions is the largest. Let's suppose that each fraction represents the number of shots we make divided by the number of attempts we make. Let's let our previous success rate be $\frac{a}{a+b}$, where a is the number of shots we made and b is the number of shots we didn't make. If we make an additional shot, it adds one to the top and bottom of this fraction, and intuitively it seems like making one shot in should improve our success rate. The fractions are arranged so that from the first to the last, the numerator and denominator both increase by one, which tells us that the last answer choice, $\frac{5}{6}$, is the largest.

2 Which of these fractions is the biggest?

Your Answer $\frac{2}{1}$

Correct Answer $\frac{2}{1}$

Explanation

This question looks very much like the previous one! However, one difference is that from the first answer choice to the last, instead of adding one to the numerator and denominator of each fraction, we are subtracting one from the numerator and denominator of each fraction. These fractions happen to be reciprocals of the fractions from the previous question, and we see that there is an inverse correlation here, since the reciprocals get bigger as we subtract more and more from the top and bottom. The answer, by inspection, is $\frac{2}{1} = 2$, as the other choices are all less than 2.

3 Which of the following decimals is equal to $\frac{3}{7}$?

Your Answer	$0.\overline{428571...}$
Correct Answer	$0.\overline{428571...}$

Explanation

Doing the long division will give you $\frac{3}{7} = 0.\overline{428571...}$, a *repeating* decimal. However, the long division doesn't have to be long, since you can just figure out the first digit by doing $\frac{30}{7} = 4 + \text{something}$. Wait; did you notice that the other choices are actually the decimal representations of similar fractions?

$$0.\overline{428571...} = \frac{3}{7}$$

$$0.\overline{285714...} = \frac{2}{7}$$

$$0.\overline{857142...} = \frac{6}{7}$$

$$0.\overline{571428...} = \frac{4}{7}$$

$$0.\overline{714285...} = \frac{5}{7}$$

What is neat is that all these similar fractions have a repeating decimal sequence of 142857 !

4

Find the value of

$$\frac{14}{15} \times \frac{12}{13} \times \frac{16}{17} \times \frac{13}{14} \times \frac{15}{16}.$$

Your answer should be a fraction in simplest form written without spaces, e.g. 1/2.

✕	**Your Answer**	[no answer given]
	Correct Answer	12/17

Explanation

If you rearrange the fractions, it becomes easier to see that a lot of terms will cancel:

$$\frac{14}{15} \times \frac{12}{13} \times \frac{16}{17} \times \frac{13}{14} \times \frac{15}{16}$$
$$= \frac{12}{13} \times \frac{14}{13} \times \frac{15}{14} \times \frac{16}{15} \times \frac{17}{16}.$$

All the numbers except for 12 and 17 cancel, leaving us with $\frac{12}{17}$. This is called a telescoping product, because the "inner" numbers are hidden, and only the "outer" numbers remain.

5 If your basketball shooting success rate is $\frac{1}{3}$, then which of the following is true?

Your Answer	You made one success shot, and three times as many total shots (misses plus successes)
Correct Answer	You made one success shot, and three times as many total shots (misses plus successes)

Explanation

Your basketball shooting success rate tells you not the relationship between the shots you made and the shots that you missed, but the relationship between the shots you made and the shots that you attempted altogether. Intuitively, we can imagine a pie chart, with 100% being the whole thing, representing the total ways.

The first choice is wrong because if you made one miss shot and three successful shots, then your success rate would be $\frac{3}{1+3} = \frac{3}{4}$.

The second choice is wrong because if you made one successful shot and three miss shots, your success rate would be $\frac{1}{1+3} = \frac{1}{4}$.

The third choice is wrong because if you made four successful shots, with no other shots, your success rate would be $\frac{4}{4} = 100\%$.

The fourth choice is wrong because your success rate would be $\frac{3}{3+1} = \frac{1}{4}$.

The fifth choice is correct, because your success rate would be $\frac{\text{successes}}{\text{total}} = \frac{1}{3}$.

6 Find which of the following is equal to

$$0.1\overline{666} \times 12$$

✕	**Your Answer**	[no answer given]
	Correct Answer	2

Explanation

We can write the decimal $.1\overline{666}$ as the fraction $\frac{1}{6}$, which turns our expression into something very recognizable:

$$\frac{1}{6} \times 12 = 2.$$

7 What is $1.125 \times 2.\overline{666}$?

✕	**Your Answer**	[no answer given]
	Correct Answer	3

Explanation

This looks a lot like the question from Day 1! For an easier time, convert the decimals into fractions. Then $1.125 \times 2.\overline{666}$ becomes

$$\frac{9}{8} \times \frac{8}{3}$$

which, by canceling 8 from the numerator and denominator, leaves us with

$$\frac{9}{3} = 3.$$

8 Find the value of

$$\left(\frac{1}{1+\frac{1}{3}}\right)\times\left(\frac{1}{1+\frac{1}{4}}\right)\times\left(\frac{1}{1+\frac{1}{5}}\right)\times\left(\frac{1}{1+\frac{1}{6}}\right)$$

Your answer should be a fraction written without spaces, like this: 1/2

✕	**Your Answer**	[no answer given]
	Correct Answer	3/7

Explanation

Looking at this, we could calculate each set of parentheses separately, but a niftier thing to do is to find a formula for any expression of the form

$$\frac{1}{1+\frac{1}{n}}$$

for some integer n. Now, the bottom looks like $1+\frac{1}{n}$, which we turn into one fraction by writing 1 as $\frac{n}{n}$. Then

$$\begin{aligned}\frac{1}{1+\frac{1}{n}} &= \frac{1}{\frac{n}{n}+\frac{1}{n}}\\ &= \frac{1}{\frac{n+1}{n}}\\ &= \frac{n}{n+1}\end{aligned}$$

which is nice. Then, we can plug $n=3,4,5,$ and 6 into our expression to get a telescoping product:

$$\begin{aligned}\left(\frac{1}{1+\frac{1}{3}}\right)\times\left(\frac{1}{1+\frac{1}{4}}\right)\times\left(\frac{1}{1+\frac{1}{5}}\right)\times\left(\frac{1}{1+\frac{1}{6}}\right) &= \frac{3}{4}\times\frac{4}{5}\times\frac{5}{6}\times\frac{6}{7}\\ &= \frac{3}{7}.\end{aligned}$$

9 While using a computer program, you accidentally scaled a photograph down to 80% of its previous size, and saved, so you can't undo your mistake. By what percent should you increase the dimensions of the photograph in order to undo your mistake?

Your Answer	25%
Correct Answer	25%

Explanation

The current photograph is $80\% = \frac{4}{5}$ of its original dimensions, so we can write

$$\text{current} = \frac{4}{5} \times \text{original}$$

We need to cancel the $\frac{4}{5}$ in order to get back the original dimensions, which means multiply both sides by the reciprocal of $\frac{4}{5}$, which is $\frac{5}{4}$.

$$\begin{aligned} \frac{5}{4} \times \text{current} &= \frac{5}{4} \times \frac{4}{5} \times \text{original} \\ &= \frac{125}{100} \times \text{original} \\ &= \text{original} \end{aligned}$$

This means we need to increase the dimensions of the current photograph by 25%.

10 Over summer you visit your grandmother, and return home to find that you've gained weight; 11% of your weight, to be precise. Shortly after, you eat some mysterious cafeteria food and fall sick for several weeks, which makes you lose 11% of your current weight. How does your weight now compare to your weight before summer?

Your Answer	Your weight now is down by 1.21%
Correct Answer	Your weight now is down by 1.21%

Explanation

Let's ask ourselves how much the weight gain compares to the weight loss. Is it the same, more, or less? The nominal percents are the same, but when you lose 11% of your weight after getting sick, the starting weight from this point is more than your starting weight before seeing grandmother. This means that more weight is subtracted than is added. To find the exact amount, we can say that your weight after seeing grandmother is

$$\text{weight} \times 1.11$$

and your weight after getting sick is the above weight, with 11% of it subtracted:

$$\begin{aligned}\text{end weight} &= \text{weight} \times 1.11 - \text{weight} \times 1.11 \times 0.11 \\ &= \text{weight} \times 1.11 \times \left(1 - 0.11\right) \\ &= \text{weight} \times 1.11 \times 0.89 \\ &= \text{weight} \times 0.9879\end{aligned}$$

As a percent, this means your weight after getting sick is 98.79% of your initial weight, so you lost 1.21% of your weight since the beginning.

Earned 4 of 7 points. 57%

Bonus Challenge Questions

These optional questions will not count towards your final score. However, we strongly suggest trying them for the benefit of the challenge!

11 Baby Alice spends $\frac{5}{8}$ of the day sleeping, $\frac{5}{24}$ of the day eating, and the rest of the day playing. Baby Bob spends $\frac{7}{12}$ of the day sleeping, $\frac{1}{6}$ of the day eating, and the rest of the day playing. How many more hours a day does Bob spend playing compared to Alice?

Your Answer	2
Correct Answer	2

Explanation

Let's find a common denominator in order to be able to compare all of these different fractions. Looking at the denominators of our fractions so far, we see that they are 8, (which is 2^3), 24, (which is 3×2^3), 5, and 6, (which is 2×3). Then our common denominator should have a prime factorization of

$$2^3 \times 3 \times 5$$

which equals 120. Baby Alice spends this fraction of a day sleeping and playing:

$$\begin{aligned} \frac{5}{8} + \frac{5}{24} &= \frac{75}{120} + \frac{25}{120} \\ &= \frac{100}{120}. \end{aligned}$$

She then spends the rest of her time, or $1 - \frac{100}{120} = \frac{20}{120}$ of the day playing.

Baby Bob spends this fraction of a day sleeping and playing:

$$\begin{aligned} \frac{7}{12} + \frac{1}{6} &= \frac{70}{120} + \frac{20}{120} \\ &= \frac{90}{120}. \end{aligned}$$

Then we know that Bob spends $1 - \frac{90}{120} = \frac{30}{120}$ of his day playing. Subtract the fraction of the day that Alice plays from the fraction of the day that Bob plays to get

$$\frac{30}{120} - \frac{20}{120} = \frac{10}{120}$$

$$= \frac{10}{120}$$

$$= \frac{1}{12}$$

of a day that Bob plays more than Alice. Since there are 24 hours in a day, this translates into 2 hours.

12

Einstein was once quoted as saying, "Half of my students study math, one-third of my students study geometry, one-seventh study chemistry, and 20 of them don't study at all." (For the purposes of this question, assume that math does not include geometry.) How many students did Einstein have?

✓	**Your Answer**	840
	Correct Answer	840

Explanation

If we find the fraction of Einstein's students that don't study, then we can figure out how many students he has. The ones who study are:

$$\frac{1}{2}+\frac{1}{3}+\frac{1}{7}$$

We can use a common denominator of $2 \times 3 \times 7$ in order to add these fractions:

$$\frac{21}{42}+\frac{14}{42}+\frac{6}{42}=\frac{41}{42}$$

which means that the fraction of Einstein's students who don't study is

$$1-\frac{41}{42}=\frac{1}{42}$$

Then $\frac{1}{42} \times \text{total} = 20$ students, which means that the total number of students is $20 \times 42 = 840$.

13 Find the last digit of $(\frac{5}{8})^5$.

Your Answer	8
Correct Answer	5

Explanation

Explanation:

This is rather difficult to do unless you convert $\frac{5}{8}$ into a decimal. Since $\frac{1}{8}$ is half of $\frac{1}{4}$, its decimal expansion is quite nice, and it terminates.

$$\frac{1}{8} = 0.125,$$

so $\frac{5}{8} = 0.625$. Now, we don't need to know the entire value of 0.625^5, only the last digit. Since 0.625 ends in a 5, and all powers of 0.625 end in a 5, our answer is 5.

14 Calculate

$$33\left(3+\frac{1}{3+\frac{1}{3+\frac{1}{3}}}\right)$$

✕	**Your Answer**	10
	Correct Answer	109

Explanation

$$33\left(3+\frac{1}{3+\frac{1}{3+\frac{1}{3}}}\right)$$

$$=33\left(3+\frac{1}{3+\frac{1}{\frac{9}{3}+\frac{1}{3}}}\right)$$

$$=33\left(3+\frac{1}{3+\frac{1}{\frac{10}{3}}}\right)$$

$$=33\left(3+\frac{1}{3+\frac{3}{10}}\right)$$

$$=33\left(3+\frac{1}{\frac{30}{10}+\frac{3}{10}}\right)$$

$$=33\left(3+\frac{1}{\frac{33}{10}}\right)$$

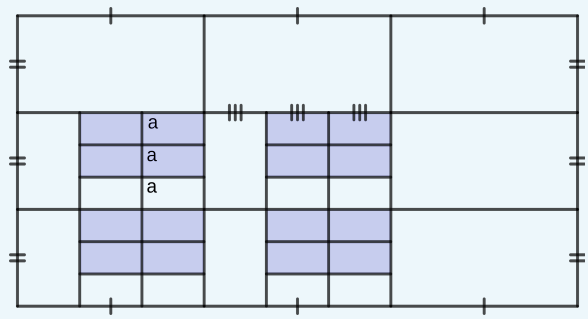
$$=33\left(3+\frac{10}{33}\right)$$

$$=33\left(\frac{99}{33}+\frac{10}{33}\right)$$

$$=33\left(\frac{109}{33}\right)$$

$$= 109$$

15



What fraction of the rectangle above is colored white? Your answer should be a fraction without spaces, e.g. 1/2.

✕	**Your Answer**	[no answer given]
	Correct Answer	65/81

Explanation

It's easier to find the fraction of this rectangle that is colored purple. Let's look at the lengths given and notice that the rectangle's length has been cut into thirds, each a rectangle with dimensions $3a \times b$, with only two of the three thirds having purple within them. Each of these thirds is cut into thirds as well, each third having dimensions of $a \times b$, and each of these rectangles is cut into thirds, each having dimension $\frac{1}{3}a \times \frac{1}{3}b$. Of these tiny rectangles, $\frac{2}{3}$ of them are colored purple. In all, it looks like we are doing series of multiplications of $\frac{2}{3}$ in order to get the fraction that is purple:

$$\frac{2}{3} \times \frac{2}{3} \times \frac{2}{3} \times \frac{2}{3}$$

$$= \frac{2^4}{3^4}$$

$$= \frac{16}{81}$$

Wait a second; this isn't our answer, is it? We want the fraction of the total area that is shaded white. To get that, we should subtract our answer from 1:

$$1 - \frac{16}{81} = \frac{65}{81}.$$

This doesn't simplify, so this is our final answer.

16 Which of the following fractions is the largest?

Your Answer $\frac{50}{101}$

Correct Answer $\frac{50}{101}$

Explanation

Did you notice that each of these fractions is close to $\frac{1}{2}$? We can write each one as $\frac{1}{2} + \text{something}$.

First choice:

$$\begin{aligned}\frac{4}{9} &= \frac{4.5 - 0.5}{9} \\ &= \frac{1}{2} - \frac{0.5}{9} \\ &= \frac{1}{2} - \frac{1}{18}.\end{aligned}$$

Second choice:

$$\begin{aligned}\frac{50}{101} &= \frac{50.5 - 0.5}{101} \\ &= \frac{1}{2} - \frac{0.5}{101} \\ &= \frac{1}{202}\end{aligned}$$

Third choice:

$$\begin{aligned}\frac{49}{99} &= \frac{49.5 - 0.5}{99} \\ &= \frac{1}{2} - \frac{0.5}{99} \\ &= \frac{1}{2} - \frac{1}{198}\end{aligned}$$

Fourth choice:

$$\begin{aligned}\frac{9}{19} &= \frac{9.5 - 0.5}{19} \\ &= \frac{1}{2} - \frac{0.5}{19} \\ &= \frac{1}{2} - \frac{1}{38}\end{aligned}$$

Fifth choice:

$$\begin{aligned}\frac{99}{200} &= \frac{100-1}{200} \\ &= \frac{100}{200} - \frac{1}{200} \\ &= \frac{1}{2} - \frac{1}{200}\end{aligned}$$

These all have the form "$\frac{1}{2}$ minus something", so the largest one will be the one where the term to be subtracted is the smallest. Let's compare all of these minus terms: $\frac{1}{18}$, $\frac{1}{38}$, $\frac{1}{198}$, $\frac{1}{200}$, and $\frac{1}{202}$. From this, we can see that $\frac{1}{202}$ is the smallest minus term, so the largest answer will be $\frac{50}{101}$.

17 Your initial basketball shot success rate is $\frac{1}{12}$, but if you make four more consecutive shots (without any misses), then your success rate will become $\frac{3}{14}$. How many misses did you make so far?

Your Answer	22
Correct Answer	22

Explanation

Explanation:

Let's think about how to convert the information in this problem into math. Let's let a mean the number of success shots you made initially, before the additional 4 success shots. Let b mean the number of miss shots you made. Since the success rate is $\frac{\text{successes}}{\text{misses }+\text{successes}}$, then

$$\frac{a}{a+b}=\frac{1}{12}.$$

Solve for b in terms of a:

$$\begin{aligned}12a&=a+b\\11a&=b\end{aligned}$$

Then, let's translate the next piece of information into more math. After 4 successful shots in a row, the success rate will become

$$\frac{a+4}{a+b+4}$$

Substituting $b=11$ into this equation, we get

$$\begin{aligned}\frac{a+4}{a+11a+4}&=\frac{3}{14}\\\frac{a+4}{12a+4}&=\frac{3}{14}\\14\left(a+4\right)&=3\left(12a+4\right)\\14a+56&=36a+12\\44&=22a\\2&=a\end{aligned}$$

This is the number of successful shots we made initially, but we want to find the number of misses originally, which is

$$b = 11a = 11 \times 2 = 22.$$

Earned 3 of 3 points. 100%

Super Bonus Questions

These questions are an extreme challenge! Some of the most interesting, creative and clever questions are the hardest ones. So try every single one. Even if you can't figure it out, you'll still be deepening your understanding of the important math concepts that underlie this course.

18 Which of the following fractions is the largest?

Your Answer	$\frac{2223}{2225}$
Correct Answer	$\frac{2223}{2225}$

Explanation

Rewrite the above fractions as

$$1-\frac{1}{1111}1-\frac{2}{2225}\ 1-\frac{3}{3337}$$

Now we can find which of $\frac{1}{1111}$, $\frac{2}{2225}$, and $\frac{3}{3337}$ is smallest. Compare the first two fractions:

$$\frac{1}{1111}=\frac{2}{2222}>\frac{2}{2225},$$

so the first fraction is not the smallest. Now consider the second and third fractions:

$$\frac{2}{2225}=\frac{6}{6675}$$

$$\frac{3}{3337}=\frac{6}{6674}$$

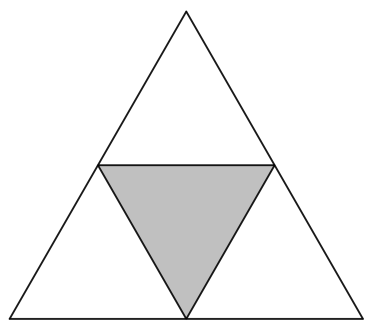
Notice that $\frac{6}{6675}<\frac{6}{6674}$, so the second fraction is smaller. Thus, $\frac{2}{2225}$ is the smallest out of $\frac{1}{1111}$, $\frac{2}{2225}$, and $\frac{3}{3337}$, so $\frac{2223}{2225}$ must be the largest out of $\frac{1110}{1111}$, $\frac{2223}{2225}$, and $\frac{3334}{3337}$.

19 Repeat the following steps, starting with an equilateral triangle: connect the midpoints of the three sides to form four smaller equilateral triangles. Color the innermost triangle black. Repeat this process with all remaining white equilateral triangles for a certain number of times, n, until the fraction of the large triangle that is shaded black is $\frac{37}{64}$. (The first step should be included in the value of n.)
What is the value of n?

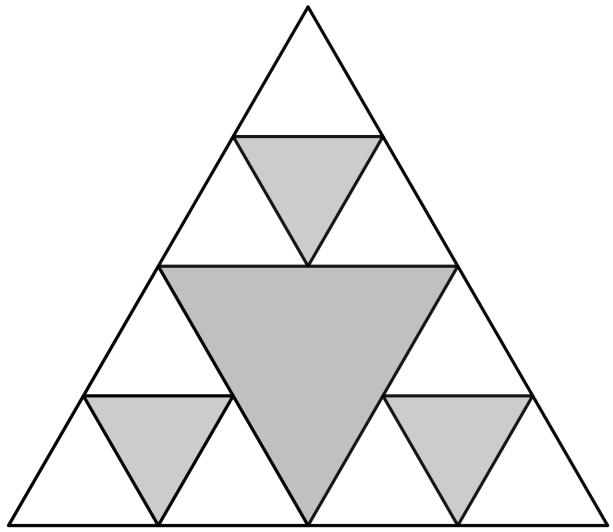
✓	**Your Answer**	3
	Correct Answer	3

Explanation

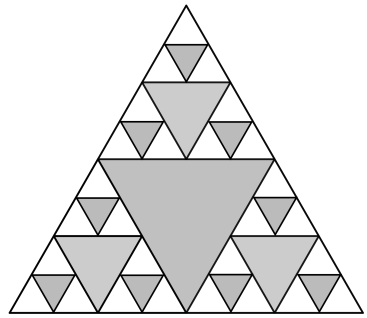
After the first coloring, we have $\frac{1}{4}$ of the large triangle colored black, and four triangles of side length $\frac{s}{2}$.



After the second step, we cut the three medium white triangles, which take up $\frac{3}{4}\text{th}$ of the total space, into $4 \times 3 = 12$ triangles of side length $\frac{s}{4}$, and color three of these triangles, so we color $\frac{3}{12} \times \frac{3}{4}$ additionally.



Next, we have 9 white triangles left, which themselves are $\frac{9}{16}$ of the total space of the large triangle (since they came from 4 medium triangles which were each divided by 4. By the same process, $\frac{1}{4}$ of each of these 9 white triangles is colored black, so we color an additional $\frac{9}{16} \times \frac{1}{4}$.



It's not clear how many times we need to repeat this process, but you can use estimation to help you. The amount of shaded triangle, $\frac{37}{64}$, is close to $\frac{1}{2}$. Though each additional coloring adds less and less black, since the areas of successive triangles is shrinking rapidly, we already get much of the way there from the first two steps, $\frac{1}{4} + \frac{3}{16}$. The second fraction is close to $\frac{1}{4}$, so the sum of the first two steps is close to half. We might guess that we are close already, so calculating the first few steps by hand is a reasonable thing to do.

$$\begin{aligned} &\frac{1}{4} + \frac{1}{4} \times \frac{3}{4} + \frac{9}{16} \times \frac{1}{4} \\ &= \frac{37}{64} \end{aligned}$$

This means that it took three coloring steps to reach the fraction of $\frac{37}{64}$ that we wanted.

20 You've made 8 liters of lemonade that is 50% water and 50% lemonade mix. However, it is much too sweet for you, and you think the concentration of lemonade mix needs to be halved to 25% for optimal flavor. The only issue is, you have no pure water! The only other liquid you have on hand is another more diluted pitcher of lemonade that is 20% lemonade mix. How many liters of the diluted lemonade must be added to your original 8 in order to get the desired concentration?

✓	**Your Answer**	40
	Correct Answer	40

Explanation

Since the original 8 liters is 50% lemonade mix, we have 4 liters of lemonade mix originally. If we add x liters of the diluted lemonade, 20% of the added volume will be lemonade mix, or $0.2x$. Since we want a 25% concentration at the end, we must solve

$$\begin{aligned} \frac{4+0.2x}{8+x} &= 0.25 \\ 4+0.2x &= 2+0.25x \\ 2 &= 0.05x \\ x &= 40. \end{aligned}$$

Summary

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